

Topic to discuss

- What is Differentiation
- What is Numerical differentiation
- Forward differentiation
- Backward differentiation
- Numerical Problem
- Homework Problem

What is Differentiation ?

→ It is a mathematical Process used to find the rate of change of one quantity with respect to another.

eg: If $y = x^2$
then its derivative, $\frac{dy}{dx} = 2x$

Simply,

Differentiation tells us,

how fast something is changing

how steep a curve is at a point

how one variable depends on another.

Example: Differentiation mean rate of change.

Suppose a car is moving on a road.

Its position changes with time.

So we can write: s = position

where s depends on time t .

Velocity = Rate of change of position wrt time

$$v = \frac{ds}{dt}, \text{ It means velocity is derivative of position.}$$

Acceleration = Rate of change of Velocity w.r.t time

$$a = \frac{dv}{dt}, \text{ It means acceleration is derivative of velocity.}$$

$$a = \frac{d}{dt} \left(\frac{ds}{dt} \right)$$

$$a = \frac{d^2s}{dt^2}; \text{ it means it is also second order derivation of position.}$$

Numerical Differentiation

Numerical differentiation is a method used to find the derivative approximately when the actual equation is not known clearly, and only some data values are given.

1. Numerical differentiation for equal interval
2. Numerical differentiation for unequal interval

x	y
x_0	y_0
x_1	y_1
x_2	y_2
x_3	y_3

Here we have to find

$y'(x_i)$ or $y''(x_i)$ for any $x = x_i$

This is called numerical differentiation.

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Newton Forward Difference :

$$y(x) = y_0 + u \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!} \Delta^3 y_0 + \frac{u(u-1)(u-2)(u-3)}{4!} \Delta^4 y_0 + \dots$$

where, $u = \frac{x-x_0}{h}$, $x = x_0 + uh$

$$\Rightarrow y(x_0 + uh) = y_0 + u \Delta y_0 + \frac{u^2 - u}{2!} \Delta^2 y_0 + \frac{u^3 - 3u^2 + 2u}{3!} \Delta^3 y_0 + \frac{u^4 - 6u^3 + 11u^2 - 6u}{4!} \Delta^4 y_0 + \dots$$

Differentiating w.r.t u , we get,

$$h y'(x_0 + uh) = \Delta y_0 + \frac{2u-1}{2!} \Delta^2 y_0 + \frac{3u^2 - 6u + 2}{3!} \Delta^3 y_0 + \frac{4u^3 - 18u^2 + 22u - 6}{4!} \Delta^4 y_0$$

$$y'(x_0 + uh) = \frac{1}{h} \left[\Delta y_0 + \frac{2u-1}{2!} \Delta^2 y_0 + \frac{3u^2 - 6u + 2}{3!} \Delta^3 y_0 + \frac{4u^3 - 18u^2 + 22u - 6}{4!} \Delta^4 y_0 + \dots \right]$$

Differentiating again w.r.t u ,

$$y''(x_0 + uh) = \frac{1}{h^2} \left[\Delta^2 y_0 + (u-1) \Delta^3 y_0 + \frac{6u^2 - 18u + 11}{12} \Delta^4 y_0 + \dots \right]$$

Newton Backward difference :

$$y(x) = y_n + u \nabla y_n + \frac{u(u+1)}{2!} \nabla^2 y_n + \frac{u(u+1)(u+2)}{3!} \nabla^3 y_n + \frac{u(u+1)(u+2)(u+3)}{4!} \nabla^4 y_n + \dots$$

where, $u = \frac{x - x_n}{h}$, $x = x_n + uh$

$$y(x_n + uh) = y_n + u \nabla y_n + \frac{u^2 + u}{2!} \nabla^2 y_n + \frac{u^3 + 3u^2 + 2u}{3!} \nabla^3 y_n + \frac{u^4 + 6u^3 + 11u^2 + 6u}{4!} \nabla^4 y_n + \dots$$

Differentiating w.r.t u , we get,

$$h y'(x_n + uh) = \nabla y_n + \frac{2u+1}{2!} \nabla^2 y_n + \frac{3u^2+6u+2}{3!} \nabla^3 y_n + \frac{4u^3+18u^2+22u+6}{4!} \nabla^4 y_n + \dots$$

$$y'(x_n + uh) = \frac{1}{h} \left[\nabla y_n + \frac{2u+1}{2!} \nabla^2 y_n + \frac{3u^2+6u+2}{3!} \nabla^3 y_n + \frac{4u^3+18u^2+22u+6}{4!} \nabla^4 y_n + \dots \right]$$

Differentiating again w.r.t u ,

$$y''(x_n + uh) = \frac{1}{h^2} \left[\nabla^2 y_n + (u+1) \nabla^3 y_n + \frac{6u^2-18u+11}{12} \nabla^4 y_n + \dots \right]$$

Numerical Problem:

Q: from the following table, find

- 1) y' at $x = 1.0$ 2) y'' at $x = 1.0$

x	1.0	1.1	1.2	1.3	1.4
y	2.7183	3.0042	3.3201	3.6693	4.0552

Solution :- Here, $h = (1.1 - 1.0)$
 $= 0.1$

Since we need derivative at the beginning point $x = 1.0$, we use Newton forward difference formula,

forward difference table,

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1.0	2.7183	0.2859	0.0300	0.0033	-0.0004
1.1	3.0042	0.3159	0.0333	0.0029	
1.2	3.3201	0.3492	0.0362		
1.3	3.6693	0.3859			
1.4	4.0552				

$$\text{So, } u = \frac{x - x_0}{h} = \frac{1.0 - 1.0}{0.1}$$

$$u = 0$$

Newton Forward difference derivatives formula,

$$y'(x_0 + uh) = \frac{1}{h} \left[\Delta y_0 + \frac{2u-1}{2!} \Delta^2 y_0 + \frac{3u^2-6u+2}{3!} \Delta^3 y_0 + \frac{4u^3-18u^2+22u-6}{4!} \Delta^4 y_0 + \dots \right]$$

$$y' = \frac{1}{h} \left[\Delta y_0 - \frac{1}{2!} \Delta^2 y_0 + \frac{1}{3} \Delta^3 y_0 - \frac{1}{4} \Delta^4 y_0 + \dots \right]$$

$$= \frac{1}{0.1} \left[0.2859 - \frac{1}{2} \times 0.0300 + \frac{1}{3} \times 0.0033 - \frac{1}{4} \times (-0.0004) \right]$$

$$= 10 (0.2859 - 0.0150 + 0.0011 + 0.0001)$$

$$= 10 \times 0.2721$$

$$= 2.721$$

$$\therefore y'(1.0) = 2.721 \text{ Ans.}$$

Now,

$$y''(x_0 + uh) = \frac{1}{h^2} \left[\Delta^2 y_0 + (u-1) \Delta^3 y_0 + \frac{6u^2 - 18u + 11}{12} \Delta^4 y_0 + \dots \right]$$

$$y'' = \frac{1}{h^2} \left[\Delta^2 y_0 - \Delta^3 y_0 + \frac{11}{12} \Delta^4 y_0 \right]$$

$$= \frac{1}{(0.1)^2} \left[0.0300 - 0.0033 + \frac{11}{12} \times (-0.0004) \right]$$

$$= 100 [0.0300 - 0.0033 - 0.0003667]$$

$$= 100 \times 0.0263333$$

$$= 2.63333$$

$$\therefore y'' \text{ at } x=1.0 = 2.63333$$

Ans.

Homework Problem:

Q: find y' and y'' at $x = 1.05$ from the table,

x	1.0	1.1	1.2	1.3	1.4
y	1.00	1.11	1.25	1.42	1.61

solution: $\therefore h = (1.1 - 1.0) = 0.1$

Since, $x = 1.05$ is not in the table and is near the beginning, we use Newton forward differentiation.

Newton forward difference table,

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1.0	1.00	0.11	0.03		
1.1	1.11	0.14	0.03	0.00	
1.2	1.25	0.17	0.02	-0.01	-0.01
1.3	1.42	0.19			
1.4	1.61				

$$\begin{aligned}\therefore u &= \frac{x - x_0}{h} \\ &= \frac{1.05 - 1.0}{0.1} = 0.5\end{aligned}$$

Now,

$$y'(x_0+uh) = \frac{1}{h} \left[\Delta y_0 + \frac{2u-1}{2!} \Delta^2 y_0 + \frac{3u^2-6u+2}{3!} \Delta^3 y_0 + \frac{4u^3-18u^2+22u-6}{4!} \Delta^4 y_0 + \dots \right]$$
$$= \frac{1}{0.1} \left[0.11 + \frac{2 \times 0.5 - 1}{2} \times 0.03 + \frac{(3 \times 0.5^2 - 6 \times 0.5 + 2)}{6} \times 0 \right. \\ \left. + \frac{4 \times 0.5^3 - 18 \times 0.5^2 + 22 \times 0.5 - 6}{24} \times -0.01 \right]$$

$$= \frac{1}{0.1} [0.11 + 0.0417 \times (-0.01)]$$

$$= \frac{1}{0.1} [0.11 - 0.000417]$$

$$= 1.0958$$

$$\therefore y'(1.05) = 1.0958$$

Ans.

Now,

$$\begin{aligned}y''(x_0 + uh) &= \frac{1}{h^2} \left[\Delta^2 y_0 + (u-1) \Delta^3 y_0 + \frac{6u^2 - 18u + 11}{12} \Delta^4 y_0 + \dots \right] \\&= \frac{1}{0.1^2} \left[0.03 + (0.5-1) \times 0 + \frac{(6 \times 0.5^2 - 18 \times 0.5 + 11)}{12} \times -0.01 \right] \\&= 100 (0.03 - 0.002917) \\&= 100 \times 0.027083 \\&= 2.7083\end{aligned}$$

$$\therefore y''(1.05) = 2.7083$$

Ans.

Homework Problem :

Q: find y' and y'' at $x=2.35$ from the following table:

x	1.95	2.05	2.15	2.25	2.35
y	3.80	4.20	4.68	5.24	5.88

solution: $h = (2.05 - 1.95)$
 $= 0.10$

Since, $x = 2.35$ is at the end of the table
we use Newton Backward formula.

Newton Backward table,

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
1.95	3.80				
2.05	4.20	0.40	0.08	0	
2.15	4.68	0.48	0.08	0	0
2.25	5.24	0.56	0.08		
2.35	5.88	0.64			

for newton backward,

$$u = \frac{x - x_n}{h} = \frac{2.35 - 2.35}{0.10}$$

$$\therefore u = 0$$

$$y'(x_n + uh) = \frac{1}{h} \left[\nabla y_n + \frac{2u+1}{2!} \nabla^2 y_n + \frac{3u^2+6u+2}{3!} \nabla^3 y_n + \frac{4u^3+18u^2+22u+6}{4!} \nabla^4 y_n + \dots \right]$$

$$y' = \frac{1}{h} \left[\nabla y_n + \frac{1}{2} \nabla^2 y_n + \frac{1}{3} \nabla^3 y_n + \frac{1}{4} \nabla^4 y_n \right]$$

$$= \frac{1}{0.10} \left[0.64 + \frac{1}{2} \times 0.08 + \frac{1}{3} \times 0 + \frac{1}{4} \times 0 \right]$$

$$= \frac{1}{0.10} [0.64 + 0.04]$$

$$= 6.80$$

$$\therefore y'(2.35) = 6.80 \text{ Ans.}$$

and y'' will be,

$$y''(x_n + uh) = \frac{1}{h^2} \left[\nabla^2 y_n + (u+1) \nabla^3 y_n + \frac{6u^2 - 18u + 11}{12} \nabla^4 y_0 + \dots \right]$$

$$y'' = \frac{1}{h^2} \left[\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12} \nabla^4 y_0 \right]$$

$$= \frac{1}{(0.10)^2} \left[0.08 + 0 + \frac{11}{12} \times 0 \right]$$

$$= 8.00$$

$$\therefore y''(2.35) = 8.00 \quad \underline{\text{Ans.}}$$

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